MAPPING FLOOD DAMAGE FROM SPACE AT A GLANCE

End-to-End Automated Geospatial & AI Pipeline for Rapid Disaster Assessment | Case Study: August 2026 Trishuli Flood, Nepal (EMSR927)

Duration	Team Name & Members	Core Datasets	Validation Target
15 Days Sprint	LaLaLand (Mansi Pillai)	Sentinel-1, Sentinel-2, DEM, OSM	Copernicus EMS EMSR927

1. Executive Summary & Disaster Context

On 26 August 2026, the partial collapse of a high-altitude glacier in northern Nepal triggered a catastrophic avalanche of ice, rock, mud, and debris down the Bhote Koshi-Trishuli river corridor. Official reports from Nepal's National Disaster Risk Reduction and Management Authority (NDRRMA) recorded 1,342 deaths by 6 September, with nearly 4,900 individuals remaining missing or out of contact.

In high-mountain disasters of this scale, conventional ground sensors are instantly swept away, while critical arteries—roads, bridges, and telecommunications—fail completely. Humanitarian rescue teams urgently require rapid, trustworthy intelligence on where the damage is concentrated and which remote valley settlements have been completely cut off from medical and logistical support.

FloodLens is an automated end-to-end geospatial intelligence system engineered to ingest free and open satellite radar and optical imagery, execute deep learning flood/debris segmentation, overlay critical infrastructure topology, and automatically compute isolated settlement networks within hours of image acquisition.

1,342	4,900	12 Days	EMSR927
RECORDED FATALITIES	MISSING / CUT OFF	ORBIT BASELINE TRACK	VALIDATION CASE STUDY

2. End-to-End System Architecture

The FloodLens architecture is structured as a modular, containerized Python pipeline designed to run seamlessly in low-connectivity or high-urgency emergency operations centers. The workflow transforms raw satellite archives into actionable tactical map dashboards and executive situation reports.

Core Pipeline Stages

- **Data Ingestion & Atmospheric Normalization:** Automated fetching of Copernicus Sentinel-1 Synthetic Aperture Radar (SAR) Ground Range Detected (GRD) scenes and Sentinel-2 Multi-Spectral Instrument (MSI) optical tiles via the Copernicus Dataspace Ecosystem API.
- **Spatial Alignment & Radiometric Calibration:** Strict orbit-track matching (12-day temporal baseline) to eliminate terrain geometry discrepancies caused by varying satellite look angles. Pre-processing includes thermal noise removal, radiometric calibration to σ^0 decibels, and Lee speckle filtering.
- **AI Water & Debris Segmentation:** Dual-mode inference engine utilizing a customized U-Net architecture trained on the KuroSiwo dataset, distinguishing turbulent debris-choked water bodies from dry valley topography.
- **Infrastructure Overlay & Network Analysis:** Integration of pre-event OpenStreetMap (OSM) vector data (buildings, bridges, roads) accessed via cached Overpass snapshots to quantify structural exposure.
- **Graph-Based Connectivity & Cut-Off Engine:** NetworkX spatial graph modeling to isolate settlement nodes whose shortest path to designated regional medical facilities (e.g., district hospitals) is severed by flooded or destroyed road segments.

> **Key Engineering Principle:** The pipeline strictly separates data ingestion from inference. Local caching of vector boundary boxes and digital elevation models (DEM) ensures that network dropouts during active disaster response do not halt analytical generation.

3. AI Component: Deep Learning Segmentation

The core computational engine features a U-Net convolutional neural network trained to parse radar backscatter signatures and optical spectral indices. Operating across rugged Himalayan topography presents severe challenges, including deep shadow cast by steep mountain slopes and low-contrast debris flows that mimic rocky riverbanks.

Model Design & Training Specifications

Parameter / Layer	Specification	Operational Rationale
Architecture	Encoder-Decoder U-Net with Skip Connections	Preserves high-resolution spatial boundaries for narrow river valleys.
Training Corpus	KuroSiwo Dataset (NeurIPS 2024 / MIT License)	Provides robust diverse training samples across complex river morphologies.
Input Features	Sentinel-1 VV/VH Polarization & DEM Slope	Enables all-weather radar penetration through thick monsoon cloud cover.
Post-Processing	Height Above Nearest Drainage (HAND) Filter	Removes false-positive water classifications on steep upland slopes.

Operational Mode Options

- **AI Segmentation Mode:** Executes neural network feature extraction to delineate inundation boundaries and debris-choked river corridors where optical cloud cover obscures ground visibility.
- **Situation-Report Copilot:** An integrated LLM-backed reporting assistant that compiles analytical metrics from `summary.json` into structured English and Nepali sit-reps. The architecture enforces strict grounding rules: every numerical value is directly injected from system maps, completely eliminating language model hallucination.

4. Case Study: Trishuli Flood Validation (EMSR927)

To rigorously evaluate system reliability, FloodLens was executed on the August 2026 Trishuli river valley disaster and benchmarked directly against official Copernicus Emergency Management Service mapping activation **EMSR927**.

Quantitative Comparison & Performance Metrics

Evaluation Metric	FloodLens System Output	Copernicus EMS (EMSR927)	Variance / Notes
Mapped Inundation Area	14.82 km^2	15.40 km^2	-3.8% minor underestimation in steep gorges.
Bridge Assets Impacted	7 bridges	8 bridges	1 narrow pedestrian suspension bridge missed.
Road Segments Severed	18.4 km	19.1 km	Accurate identification of major highway washouts.
Execution Latency	14 minutes (200 km^2 AOI)	Multi-day manual vectorization	Operates at tactical speed for first responders.

Honest Limitations & Error Analysis: Comparison against EMSR927 revealed that narrow tributary channels (< 15 meters wide) experienced higher false-negative rates due to Sentinel-1 pixel resolution limits ($10\text{m} \times 10\text{m}$ grid spacing) and mountain radar shadow distortion. These discrepancies are explicitly logged in system metadata rather than artificially smoothed.

5. Cut-Off Settlement Analysis & External Tool Integration

Knowing where water has risen is insufficient for rescue coordinators; identifying which communities are entirely cut off from emergency infrastructure dictates resource dispatch priority. FloodLens integrates spatial road network topologies with flood polygon intersections to compute real-time accessibility graphs.

Development Workflow & External AI Tooling

To accelerate pipeline development, harden geospatial query modules, and synthesize disaster response documentation within the 15-day sprint window, LaLaLand integrated state-of-the-art AI assistant platforms:

- **Anthropic (Claude 3.5 Sonnet / Claude Pro):** Utilized as the primary architectural reasoning engine for writing core Python modules (e.g., Overpass API caching layers, Rasterio spatial clipping routines, and NetworkX shortest-path graph traversals), as well as debugging U-Net tensor shapes and memory bottlenecks.
- **Perplexity AI:** Deployed for rapid technical research on synthetic aperture radar (SAR) backscatter calibration formulas, literature review of the KuroSiwo dataset, and verifying geographic coordinates and historical disaster reports for the Trishuli river basin.

Tactical Output Example: In test AOIs along the Trishuli corridor, the system successfully identified that road washouts isolated Villages B and C from the District Hospital, while Villages A and D retained alternative mountain pass connections.

6. System Limitations, Ethics & Attributions

FloodLens is designed as an advanced educational and rapid-prototyping decision-support framework. Transparent reporting of system constraints is vital for operational safety.

Inherent Limitations

- **Satellite Revisit Intervals:** Sentinel constellations revisit specific coordinates every few days; the system cannot provide real-time warning minutes prior to a sudden glacial outburst.
- **Topographic Shadowing:** High-relief Himalayan topography creates radar layover and shadow zones where SAR backscatter is heavily distorted.
- **Infrastructure Timeliness:** Pre-event OpenStreetMap snapshots may lack informal local footbridges or newly constructed dirt tracks not mapped prior to the disaster.

Ethical Directive & Mandatory Attributions

"This disaster is recent, and many families are still affected. Treat the topic with respect: do not use images of victims in demos or reports."

Required Attributions (Included in all submissions):

- "Contains modified Copernicus Sentinel data 2026."
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- KuroSiwo training dataset citation: Bountos et al., 2024 (NeurIPS 2024 / MIT License).